

0.1 三角函数相关

0.1.1 三角函数

定理 0.1 (两角和差公式)

- (1) $\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$;
- (2) $\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$;
- (3) $\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}$;
- (4) $\cot(\alpha \pm \beta) = \frac{\cot \alpha \cot \beta \mp 1}{\cot \beta \pm \cot \alpha}$.



证明 Show/Hide Proof

- (1) 在平面直角坐标系中考虑单位圆, 设角 α 和角 β 的终边分别与单位圆交于点 P_1 和 P_2 , 则它们的坐标分别为

$$P_1 = (\cos \alpha, \sin \alpha), \quad P_2 = (\cos \beta, \sin \beta).$$

根据向量数量积的坐标表示与几何定义 (夹角为 $\alpha - \beta$), 有

$$\cos(\alpha - \beta) = P_1 \cdot P_2 = \cos \alpha \cos \beta + \sin \alpha \sin \beta.$$

将上式中的 β 替换为 $-\beta$, 并利用三角函数的奇偶性 $\sin(-\beta) = -\sin \beta$ 和 $\cos(-\beta) = \cos \beta$, 可得

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta.$$

将以上两式合并, 即得

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta.$$

- (2) 利用诱导公式 $\sin \theta = \cos\left(\frac{\pi}{2} - \theta\right)$, 可得

$$\begin{aligned} \sin(\alpha + \beta) &= \cos\left[\frac{\pi}{2} - (\alpha + \beta)\right] = \cos\left[\left(\frac{\pi}{2} - \alpha\right) - \beta\right] \\ &= \cos\left(\frac{\pi}{2} - \alpha\right) \cos \beta + \sin\left(\frac{\pi}{2} - \alpha\right) \sin \beta \\ &= \sin \alpha \cos \beta + \cos \alpha \sin \beta. \end{aligned}$$

同样用 $-\beta$ 替换 β , 得到

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta.$$

合并为

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta.$$

- (3) 根据正切与正弦、余弦的商数关系 $\tan \theta = \frac{\sin \theta}{\cos \theta}$, 代入上述公式可得

$$\tan(\alpha + \beta) = \frac{\sin(\alpha + \beta)}{\cos(\alpha + \beta)} = \frac{\sin \alpha \cos \beta + \cos \alpha \sin \beta}{\cos \alpha \cos \beta - \sin \alpha \sin \beta}.$$

将分子分母同时除以 $\cos \alpha \cos \beta$, 得

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}.$$

将 β 替换为 $-\beta$, 由正切函数的奇偶性 $\tan(-\beta) = -\tan \beta$, 可得

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}.$$

结合以上两式, 即得

$$\tan(\alpha \pm \beta) = \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta}.$$

(4) 根据余切与正切的倒数关系, 并结合 $\tan$ 的和差公式, 将分子分母同时除以 $\tan \alpha \tan \beta$ 进行化简:

$$\cot(\alpha + \beta) = \frac{1}{\tan(\alpha + \beta)} = \frac{1 - \tan \alpha \tan \beta}{\tan \alpha + \tan \beta} = \frac{\frac{1}{\tan \alpha \tan \beta} - 1}{\frac{1}{\tan \beta} + \frac{1}{\tan \alpha}} = \frac{\cot \alpha \cot \beta - 1}{\cot \beta + \cot \alpha}.$$

同样地, 可得

$$\cot(\alpha - \beta) = \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha}.$$

合并后即得到

$$\cot(\alpha \pm \beta) = \frac{\cot \alpha \cot \beta \mp 1}{\cot \beta \pm \cot \alpha}.$$

□

定理 0.2 (倍角公式)

- (1) $\sin 2\theta = 2 \sin \theta \cos \theta$;
- (2) $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$;
- (3) $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$;
- (4) $\cot 2\theta = \frac{\cot^2 \theta - 1}{2 \cot \theta}$;
- (5) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$;
- (6) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$;
- (7) $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$;
- (8) $\cot 3\theta = \frac{\cot^3 \theta - 3 \cot \theta}{3 \cot^2 \theta - 1}$;
- (9) $\sin 4\theta = 4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta)$;
- (10) $\cos 4\theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$.



证明 Show/Hide Proof

(1) 根据两角和的正弦公式 $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$, 令 $\alpha = \beta = \theta$, 可得

$$\sin 2\theta = \sin \theta \cos \theta + \cos \theta \sin \theta = 2 \sin \theta \cos \theta.$$

(2) 根据两角和的余弦公式 $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$, 令 $\alpha = \beta = \theta$, 可得

$$\cos 2\theta = \cos \theta \cos \theta - \sin \theta \sin \theta = \cos^2 \theta - \sin^2 \theta.$$

再结合同角三角函数关系式 $\sin^2 \theta + \cos^2 \theta = 1$, 亦可推出

$$\cos 2\theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta.$$

(3) 根据两角和的正切公式 $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$, 令 $\alpha = \beta = \theta$, 可得

$$\tan 2\theta = \frac{\tan \theta + \tan \theta}{1 - \tan \theta \cdot \tan \theta} = \frac{2 \tan \theta}{1 - \tan^2 \theta}.$$

(4) 由正切与余切的倒数关系可知:

$$\cot 2\theta = \frac{1}{\tan 2\theta} = \frac{1 - \tan^2 \theta}{2 \tan \theta}.$$

将其分子分母同时乘以 $\cot^2 \theta$, 即可得到

$$\cot 2\theta = \frac{\cot^2 \theta - 1}{2 \cot \theta}.$$

(5) 根据两角和公式 $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$, 令 $\alpha = 2\theta, \beta = \theta$, 可得

$$\begin{aligned} \sin 3\theta &= \sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\ &= 2 \sin \theta \cos^2 \theta + (1 - 2 \sin^2 \theta) \sin \theta \\ &= 2 \sin \theta (1 - \sin^2 \theta) + \sin \theta - 2 \sin^3 \theta \end{aligned}$$

$$= 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta = 3 \sin \theta - 4 \sin^3 \theta.$$

(6) 根据两角和公式 $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$, 令 $\alpha = 2\theta, \beta = \theta$, 可得

$$\begin{aligned}\cos 3\theta &= \cos(2\theta + \theta) = \cos 2\theta \cos \theta - \sin 2\theta \sin \theta \\ &= (2 \cos^2 \theta - 1) \cos \theta - 2 \sin^2 \theta \cos \theta \\ &= 2 \cos^3 \theta - \cos \theta - 2(1 - \cos^2 \theta) \cos \theta \\ &= 2 \cos^3 \theta - \cos \theta - 2 \cos \theta + 2 \cos^3 \theta = 4 \cos^3 \theta - 3 \cos \theta.\end{aligned}$$

(7) 由两角和的正切公式 $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$, 令 $\alpha = 2\theta, \beta = \theta$, 并设 $t = \tan \theta$, 可得

$$\begin{aligned}\tan 3\theta &= \tan(2\theta + \theta) = \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} \\ &= \frac{\frac{2t}{1-t^2} + t}{1 - \frac{2t}{1-t^2} \cdot t} = \frac{\frac{2t+t-t^3}{1-t^2}}{\frac{1-t^2-2t^2}{1-t^2}} = \frac{3t-t^3}{1-3t^2} = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.\end{aligned}$$

(8) 由余切与正切的倒数关系, 并在 $\tan 3\theta$ 的表达式中将分子分母同时乘以 $\cot^3 \theta$, 可得

$$\cot 3\theta = \frac{1}{\tan 3\theta} = \frac{1 - 3 \tan^2 \theta}{3 \tan \theta - \tan^3 \theta} = \frac{\cot^3 \theta - 3 \cot \theta}{3 \cot^2 \theta - 1}.$$

(9) 利用二倍角公式 $\sin 2\theta = 2 \sin \theta \cos \theta$ 和 $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, 迭代可得

$$\sin 4\theta = 2 \sin 2\theta \cos 2\theta = 4 \sin \theta \cos \theta (\cos^2 \theta - \sin^2 \theta).$$

(10) 利用二倍角公式 $\cos 2\theta = 2 \cos^2 \theta - 1$, 迭代可得

$$\begin{aligned}\cos 4\theta &= 2 \cos^2 2\theta - 1 = 2(2 \cos^2 \theta - 1)^2 - 1 \\ &= 2(4 \cos^4 \theta - 4 \cos^2 \theta + 1) - 1 = 8 \cos^4 \theta - 8 \cos^2 \theta + 1.\end{aligned}$$

(11) 由二倍角公式 $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$ 可得

$$\begin{aligned}\cot x - 2 \cot 2x &= \frac{\cos x}{\sin x} - 2 \cdot \frac{\cos^2 x - \sin^2 x}{2 \sin x \cos x} = \frac{\cos x}{\sin x} - \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} \\ &= \frac{\cos^2 x - (\cos^2 x - \sin^2 x)}{\sin x \cos x} = \frac{\sin^2 x}{\sin x \cos x} = \tan x.\end{aligned}$$

□

定理 0.3 (三角平方差公式)

$$\sin^2 x - \sin^2 y = \sin(x-y) \sin(x+y) = \cos(y-x) \cos(y+x) = \cos^2 y - \cos^2 x.$$



证明 Show/Hide Proof

首先, 我们有

$$\cos^2 x - \cos^2 y = 1 - \sin^2 x - (1 - \sin^2 y) = \sin^2 y - \sin^2 x.$$

接着, 我们有

$$\begin{aligned}\sin(x-y) \sin(x+y) &= (\sin x \cos y - \cos x \sin y)(\sin x \cos y + \cos x \sin y) \\ &= \sin^2 x \cos^2 y - \cos^2 x \sin^2 y \\ &= \sin^2 x (1 - \sin^2 y) - (1 - \sin^2 x) \sin^2 y \\ &= \sin^2 x - \sin^2 y;\end{aligned}$$

$$\begin{aligned}\cos(y-x) \cos(y+x) &= (\cos x \cos y + \sin x \sin y)(\cos x \cos y - \sin x \sin y) \\ &= \cos^2 x \cos^2 y - \sin^2 x \sin^2 y \\ &= \cos^2 x \cos^2 y - (1 - \cos^2 x)(1 - \cos^2 y) \\ &= \cos^2 x - \cos^2 y.\end{aligned}$$

故结论得证. □

定理 0.4

1. 对 $\forall n \in \mathbb{N}$, 有

$$\begin{aligned}\sin nx &= \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} (-1)^k \binom{n}{2k+1} (\cos x)^{n-2k-1} (\sin x)^{2k+1}, \\ \cos nx &= \sum_{k=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^k \binom{n}{2k} (\cos x)^{n-2k} (\sin x)^{2k}, \\ \tan nx &= \frac{\sin nx}{\cos nx} = \frac{\sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} (-1)^k \binom{n}{2k+1} \tan^{2k+1} x}{\sum_{r=0}^{\lfloor \frac{n}{2} \rfloor} (-1)^r \binom{n}{2r} \tan^{2r} x}.\end{aligned}$$

2. 对 $\forall k \in \mathbb{N}_+$, 有

$$\begin{aligned}\sin^{2k-1} x &= \frac{1}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \sin((2k-2i-1)x), \\ \sin^{2k} x &= \frac{1}{2^{2k-1}} \sum_{i=0}^{k-1} (-1)^{k+i} \binom{2k}{i} \cos((2k-2i)x) + \frac{1}{2^{2k}} \binom{2k}{k}, \\ \cos^{2k-1} x &= \frac{1}{2^{2k-2}} \sum_{i=0}^{k-1} \binom{2k-1}{i} \cos((2k-2i-1)x), \\ \cos^{2k} x &= \frac{1}{2^{2k-1}} \sum_{i=0}^{k-1} \binom{2k}{i} \cos((2k-2i)x) + \frac{1}{2^{2k}} \binom{2k}{k}.\end{aligned}$$



注 上述结论表明: $\cos^n x$ 可以表示为 $1, \cos x, \dots, \cos nx$ 的线性组合.

证明 [Show/Hide Proof](#)

具体证明见 [Expansions of \$\sin\(nx\)\$ and \$\cos\(nx\)\$](#) .

□

0.1.2 反三角函数

定理 0.5 (常用反三角函数性质)

1.

$$\arcsin x + \arcsin y = \begin{cases} \arcsin(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & , xy < 0 \text{ 或 } x^2 + y^2 \leq 1 \\ \pi - \arcsin(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & , x > 0, y > 0, x^2 + y^2 > 1 \\ -\pi - \arcsin(x\sqrt{1-y^2} + y\sqrt{1-x^2}) & , x < 0, y < 0, x^2 + y^2 > 1 \end{cases}$$

2.

$$\arcsin x - \arcsin y = \begin{cases} \arcsin(x\sqrt{1-y^2} - y\sqrt{1-x^2}) & , xy \geq 0 \text{ 或 } x^2 + y^2 \leq 1 \\ \pi - \arcsin(x\sqrt{1-y^2} - y\sqrt{1-x^2}) & , x > 0, y < 0, x^2 + y^2 > 1 \\ -\pi - \arcsin(x\sqrt{1-y^2} - y\sqrt{1-x^2}) & , x < 0, y > 0, x^2 + y^2 > 1 \end{cases}$$

3.

$$\arccos x + \arccos y = \begin{cases} \arccos(xy - \sqrt{1-x^2}\sqrt{1-y^2}) & , x+y \geq 0 \\ 2\pi - \arccos(xy - \sqrt{1-x^2}\sqrt{1-y^2}) & , x+y < 0 \end{cases}.$$

4.

$$\arccos x - \arccos y = \begin{cases} -\arccos(xy + \sqrt{1-x^2}\sqrt{1-y^2}) & , x \geq y \\ \arccos(xy + \sqrt{1-x^2}\sqrt{1-y^2}) & , x < y \end{cases}.$$

5.

$$\arctan x + \arctan y = \begin{cases} \arctan \frac{x+y}{1-xy} & , xy < 1 \\ \pi + \arctan \frac{x+y}{1-xy}, x > 0 & , xy > 1 \\ -\pi + \arctan \frac{x+y}{1-xy}, x < 0 & , xy > 1 \end{cases}.$$

6.

$$\arctan x - \arctan y = \begin{cases} \arctan \frac{x-y}{1+xy} & , xy > -1 \\ \pi + \arctan \frac{x-y}{1+xy} & , x > 0, xy < -1 \\ -\pi + \arctan \frac{x-y}{1+xy} & , x < 0, xy < -1 \end{cases}.$$

7.

$$2 \arcsin x = \begin{cases} \arcsin(2x\sqrt{1-x^2}) & , |x| \leq \frac{\sqrt{2}}{2} \\ \pi - \arcsin(2x\sqrt{1-x^2}) & , \frac{\sqrt{2}}{2} < x \leq 1 \\ -\pi - \arcsin(2x\sqrt{1-x^2}) & , -1 \leq x < -\frac{\sqrt{2}}{2} \end{cases}.$$

8.

$$2 \arccos x = \begin{cases} \arccos(2x^2 - 1) & , 0 \leq x \leq 1 \\ 2\pi - \arccos(2x^2 - 1) & , -1 \leq x < 0 \end{cases}.$$

9.

$$2 \arctan x = \begin{cases} \arctan \frac{2x}{1-x^2}, |x| \leq 1 \\ \pi + \arctan \frac{2x}{1-x^2} & , |x| > 1 \\ -\pi + \arctan \frac{2x}{1-x^2} & , x < -1 \end{cases}.$$

10.

$$\cos(n \arccos x) = \frac{(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n}{2} \quad (n \geq 1).$$



证明 Show/Hide Proof



命题 0.1

$$\arctan x + \arctan \frac{1}{x} = \begin{cases} \frac{\pi}{2}, & x > 0 \\ -\frac{\pi}{2}, & x < 0 \end{cases}.$$



证明 Show/Hide Proof

令 $f(x) = \arctan x + \arctan \frac{1}{x}$, 则

$$f'(x) = \frac{1}{x^2+1} + \frac{1}{(\frac{1}{x})^2+1}(-\frac{1}{x^2}) = \frac{1}{x^2+1} - \frac{1}{x^2+1} = 0$$

故 $f(x)$ 为常函数, 于是就有 $f(x) = f(1) = \frac{\pi}{2}, \forall x > 0; f(x) = f(-1) = -\frac{\pi}{2}, \forall x < 0$.

□

0.1.3 双曲三角函数

命题 0.2

$$(1) \cosh x = \frac{e^x + e^{-x}}{2} \geq 1,$$

$$(2) \sinh x = \frac{e^x - e^{-x}}{2} \geq x.$$

♣

证明 Show/Hide Proof

可以分别利用均值不等式和求导进行证明.

□

命题 0.3

$$1. \cosh^2 x - \sinh^2 x = 1.$$

$$2. \cosh(2x) = 2 \cosh^2 x - 1 = 1 - 2 \sinh^2 x = \cosh^2 x + \sinh^2 x.$$

$$3. \sinh(2x) = 2 \sinh x \cosh x.$$

♣

证明 Show/Hide Proof

□